

Kiran Swamy – 2520090035

DBMS practical 2

```
mysql> USE bankdb;
Database changed
mysql> CREATE TABLE student_marks1 (
->     roll_no INT PRIMARY KEY,
->     name VARCHAR(50),
->     subject VARCHAR(50),
->     marks DECIMAL(5,2)
-> );
Query OK, 0 rows affected (0.05 sec)
```

```
mysql> DESC student_marks1;
```

Field	Type	Null	Key	Default	Extra
roll_no	int	NO	PRI	NULL	
name	varchar(50)	YES		NULL	
subject	varchar(50)	YES		NULL	
marks	decimal(5,2)	YES		NULL	

4 rows in set (0.01 sec)

```
mysql> INSERT INTO student_marks1
-> (roll_no, name, subject, marks)
-> VALUES
-> (1, 'Ravi', 'Math', 85.50),
-> (2, 'Sita', 'Math', 92.75),
-> (3, 'Anil', 'Math', 78.40),
-> (4, 'Priya', 'Math', 88.90),
-> (5, 'Vijay', 'Math', 80.25),
-> (6, 'subbusir', 'aws', 98.50),
-> (7, 'dwaraka', 'dbms', 95.75),
-> (8, 'ranjani', 'english', 97.40),
-> (9, 'kavithamam', 'aws1', 99.90),
-> (10, 'seetha', 'azure', 82.25);
Query OK, 10 rows affected (0.02 sec)
Records: 10  Duplicates: 0  Warnings: 0
```

```
mysql> SELECT * FROM student_marks1;
```

roll_no	name	subject	marks
1	Ravi	Math	85.50
2	Sita	Math	92.75
3	Anil	Math	78.40
4	Priya	Math	88.90
5	Vijay	Math	80.25
6	subbusir	aws	98.50
7	dwaraka	dbms	95.75
8	ranjani	english	97.40
9	kavithamam	aws1	99.90
10	seetha	azure	82.25

10 rows in set (0.00 sec)

```
mysql> SELECT * FROM student_marks1;
```

roll_no	name	subject	marks
1	Ravi	Math	85.50
2	Sita	Math	92.75
3	Anil	Math	78.40
4	Priya	Math	88.90
5	Vijay	Math	80.25
6	subbusir	aws	98.50
7	dwaraka	dbms	95.75
8	ranjani	english	97.40
9	kavithamam	aws1	99.90
10	seetha	azure	82.25

```
10 rows in set (0.00 sec)
```

```
mysql> SELECT COUNT(*) AS total_students  
-> FROM student_marks1;
```

total_students
10

```
1 row in set (0.02 sec)
```

```
mysql> SELECT SUM(marks)  
-> FROM student_marks1;
```

SUM(marks)
899.60

```
1 row in set (0.00 sec)
```

```
mysql> SELECT AVG(marks)
-> FROM student_marks1;
+-----+
| AVG(marks) |
+-----+
| 89.960000 |
+-----+
1 row in set (0.00 sec)
```

```
mysql> SELECT MAX(marks)
-> FROM student_marks1;
+-----+
| MAX(marks) |
+-----+
| 99.90 |
+-----+
1 row in set (0.00 sec)
```

```
mysql> SELECT MIN(marks)
-> FROM student_marks1;
+-----+
| MIN(marks) |
+-----+
| 78.40 |
+-----+
1 row in set (0.00 sec)
```

```
mysql> SELECT * FROM student_marks1
-> WHERE marks > 85;
+-----+-----+-----+-----+
| roll_no | name      | subject | marks |
+-----+-----+-----+-----+
| 1 | Ravi      | Math    | 85.50 |
| 2 | Sita      | Math    | 92.75 |
| 4 | Priya     | Math    | 88.90 |
| 6 | subbusir  | aws     | 98.50 |
| 7 | dwaraka   | dbms    | 95.75 |
| 8 | ranjani   | english | 97.40 |
| 9 | kavithamam | aws1    | 99.90 |
+-----+-----+-----+-----+
7 rows in set (0.01 sec)
```

```
mysql> SELECT * FROM student_marks1
-> WHERE marks >= 90;
```

roll_no	name	subject	marks
2	Sita	Math	92.75
6	subbusir	aws	98.50
7	dwaraka	dbms	95.75
8	ranjani	english	97.40
9	kavithamam	aws1	99.90

```
5 rows in set (0.00 sec)
```

```
mysql> SELECT * FROM student_marks1
-> WHERE marks < 80;
```

roll_no	name	subject	marks
3	Anil	Math	78.40

```
1 row in set (0.00 sec)
```

```
mysql> SELECT * FROM student_marks1
-> WHERE marks BETWEEN 80 AND 90;
```

roll_no	name	subject	marks
1	Ravi	Math	85.50
4	Priya	Math	88.90
5	Vijay	Math	80.25
10	seetha	azure	82.25

```
4 rows in set (0.00 sec)
```

```
mysql> SELECT * FROM student_marks1
-> WHERE name LIKE 'P%';
```

roll_no	name	subject	marks
4	Priya	Math	88.90

```
1 row in set (0.00 sec)
```

```
mysql> SELECT * FROM student_marks1
-> WHERE name IN ('Ravi','Sita','Vijay');
```

roll_no	name	subject	marks
1	Ravi	Math	85.50
2	Sita	Math	92.75
5	Vijay	Math	80.25

```
3 rows in set (0.00 sec)
```

```
mysql> SELECT * FROM student_marks1
-> WHERE marks > 85
-> AND (subject='Math' OR name LIKE 'P%');
```

roll_no	name	subject	marks
1	Ravi	Math	85.50
2	Sita	Math	92.75
4	Priya	Math	88.90

```
3 rows in set (0.00 sec)
```

```
mysql> UPDATE student_marks1
-> SET marks = 90.00
-> WHERE roll_no = 3;
Query OK, 1 row affected (0.01 sec)
Rows matched: 1  Changed: 1  Warnings: 0

mysql> SELECT * FROM student_marks1
-> WHERE roll_no = 3;
+-----+-----+-----+-----+
| roll_no | name | subject | marks |
+-----+-----+-----+-----+
|      3 | Anil | Math    | 90.00 |
+-----+-----+-----+-----+
1 row in set (0.00 sec)
```

```
mysql> UPDATE student_marks1
-> SET subject = 'Remedial Math'
-> WHERE marks < 80;
Query OK, 0 rows affected (0.00 sec)
Rows matched: 0  Changed: 0  Warnings: 0

mysql> SELECT * FROM student_marks1
-> WHERE subject = 'Remedial Math';
Empty set (0.00 sec)
```

```
mysql> DELETE FROM student_marks1
-> WHERE roll_no = 5;
Query OK, 1 row affected (0.01 sec)

mysql> ^C
mysql> SELECT * FROM student_marks1
-> WHERE roll_no = 5;
Empty set (0.00 sec)
```

```
mysql> DELETE FROM student_marks1
      -> WHERE marks < 75;
Query OK, 0 rows affected (0.01 sec)
```

```
mysql> SELECT * FROM student_marks1
      -> WHERE marks < 75;
Empty set (0.00 sec)
```

```
mysql> SELECT * FROM student_marks1
      -> ORDER BY marks ASC;
```

roll_no	name	subject	marks
10	seetha	azure	82.25
1	Ravi	Math	85.50
4	Priya	Math	88.90
3	Anil	Math	90.00
2	Sita	Math	92.75
7	dwaraka	dbms	95.75
8	ranjani	english	97.40
6	subbusir	aws	98.50
9	kavithamam	aws1	99.90

```
9 rows in set (0.00 sec)
```

```
mysql> SELECT * FROM student_marks1
      -> ORDER BY marks DESC;
```

roll_no	name	subject	marks
9	kavithamam	aws1	99.90
6	subbusir	aws	98.50
8	ranjani	english	97.40
7	dwaraka	dbms	95.75
2	Sita	Math	92.75
3	Anil	Math	90.00
4	Priya	Math	88.90
1	Ravi	Math	85.50
10	seetha	azure	82.25

```
9 rows in set (0.00 sec)
```

```
mysql> SELECT * FROM student_marks1
-> ORDER BY name ASC;
```

roll_no	name	subject	marks
3	Anil	Math	90.00
7	dwaraka	dbms	95.75
9	kavithamam	aws1	99.90
4	Priya	Math	88.90
8	ranjani	english	97.40
1	Ravi	Math	85.50
10	seetha	azure	82.25
2	Sita	Math	92.75
6	subbusir	aws	98.50

9 rows in set (0.00 sec)

```
mysql> SELECT * FROM student_marks1
-> ORDER BY name DESC;
```

roll_no	name	subject	marks
6	subbusir	aws	98.50
2	Sita	Math	92.75
10	seetha	azure	82.25
1	Ravi	Math	85.50
8	ranjani	english	97.40
4	Priya	Math	88.90
9	kavithamam	aws1	99.90
7	dwaraka	dbms	95.75
3	Anil	Math	90.00

9 rows in set (0.00 sec)


```
mysql> SELECT subject, SUM(marks) AS total_marks
-> FROM student_marks1
-> GROUP BY subject;
```

```
+-----+-----+
| subject | total_marks |
+-----+-----+
| Math    | 357.15      |
| aws     | 98.50       |
| dbms    | 95.75       |
| english | 97.40       |
| aws1    | 99.90       |
| azure   | 82.25       |
+-----+-----+
6 rows in set (0.00 sec)
```

```
mysql> SELECT subject, AVG(marks) AS avg_marks
-> FROM student_marks1
-> GROUP BY subject;
```

```
+-----+-----+
| subject | avg_marks |
+-----+-----+
| Math    | 89.287500 |
| aws     | 98.500000 |
| dbms    | 95.750000 |
| english | 97.400000 |
| aws1    | 99.900000 |
| azure   | 82.250000 |
+-----+-----+
6 rows in set (0.01 sec)
```

```
mysql> SELECT subject, COUNT(*) AS num_students
-> FROM student_marks1
-> GROUP BY subject;
```

```
+-----+-----+
| subject | num_students |
+-----+-----+
| Math    | 4            |
| aws     | 1            |
| dbms    | 1            |
| english | 1            |
| aws1    | 1            |
| azure   | 1            |
+-----+-----+
6 rows in set (0.00 sec)
```

```
mysql> SELECT subject, AVG(marks) AS avg_marks
-> FROM student_marks1
-> GROUP BY subject
-> HAVING AVG(marks) > 90;
```

subject	avg_marks
aws	98.500000
dbms	95.750000
english	97.400000
aws1	99.900000

4 rows in set (0.00 sec)

```
mysql> SELECT subject, COUNT(*) AS num_students
-> FROM student_marks1
-> GROUP BY subject
-> HAVING COUNT(*) > 1;
```

subject	num_students
Math	4

1 row in set (0.00 sec)